

MATH 380 Intro. to Optimization for Data Science

Section 01 Fall 2026

- Instructor: Dr. Jinglai Shen
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- Office: Math/Psyc 417
- Phone: (410) 455-2402
- Lectures: Tue and Thu, 1:00–2:15 pm, at MP 008
- Office hours: Tue and Thu, 12:00–1:00 pm
- Prerequisites: linear algebra, multi-variable calculus, basic real analysis and probability; MATLAB, Python or other numerical software programming skills are needed.
- Recommended textbooks (but not required):
 - E. Chong, W.-S. Lu, and S. H. Zak. *An Introduction to Optimization with Applications to Machine Learning*, 5th Edition, John Wiley & Sons, Inc, 2024;
 - A. Beck and N. Guttman-Beck. *A First Course in Linear Optimization*, SIAM Publisher, 2025.
- Other reference:
 - S. Roch. *Mathematical Methods in Data Science*, Cambridge University Press, 2026.
- Important dates:
 - No class on Oct. 13 (Fall Recess) and Nov. 26 (Thanksgiving)
 - Last class: Dec. 8th, Tuesday
 - Final exam: 1:00-3:00 pm, Dec. 15, Tuesday

Course Description Optimization techniques play a crucial and increasingly important role in data science, machine learning, and AI. This course aims to provide an introduction to fundamental optimization theory and algorithms that find numerous applications in data analysis, statistics, and machine learning. We will focus on first-order and second-order algorithms for a broad class of representative continuous optimization problems arising from data science; optimality conditions and convergence properties of optimization algorithms will be developed. Specific topics include unconstrained optimization, gradient methods and extensions, Newton method and variations, linear programming, simplex method, duality theory, non-simplex methods, constrained/convex optimization, projection methods, and Lagrangian methods. This course will also help you gain computational experience by writing codes and testing numerical programs in homework.

Specific topics are:

- Unconstrained Optimization
 - First-order and second-order optimality conditions
 - Gradient descent methods, convergence analysis
 - Steepest descent method, convergence rate
 - Stochastic gradient methods, accelerated methods
 - Newton method
 - Conjugate gradient methods, Quasi-Newton methods
- Linear Programming/Optimization
 - Formulation, convexity, and fundamental theorem
 - Simplex methods
 - Duality theory, dual method
 - Non-simplex methods
- Nonlinear Constrained Optimization
 - First-order and second-order optimality conditions, Lagrangian multipliers
 - Equality or inequality constrained optimization
 - Convex optimization
 - Gradient projection method, Lagrangian algorithms
- Applications to statistics, machine learning, and data science

Please note that these topics are subject to change, depending on class progress.

Homework Biweekly homework will be assigned; each homework assignment contains theoretical problems and numerical problems. Familiarity with numerical softwares, e.g., MATLAB or Python, is required for solving numerical problems. Homework must be turned in by a specified date; *no late due will be accepted*. Please present your answers neatly and show all your work; answers without supporting work may not receive full credit.

Exams There will be two (2) in-class mid-term exams and one (1) final exam. An exam date will be announced at least one week ahead. Each mid-term exam mainly focuses on topics covered up to that point in the course, whereas the final exam is comprehensive with greater emphasis on the last part of the course. Please note that there will be no optional final exam.

Grading Policy The grading scheme is as follows:

- Homework: 30%
- Mid-term exams: 44% (22% for each)
- Final exam: 26%

The letter grade will be computed based upon the numerical grade:

$A : \geq 90$; $B : 89 - 80$; $C : 79 - 65$; $D : 64 - 50$; $F : < 50$

Academic Integrity

- The UMBC Academic Integrity Statement:

“Academic integrity is an important value at UMBC. By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC’s scholarly community in which everyone’s academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal. These principles and policies apply in both face-to-face and online classes.”

To read the full Student Academic Conduct Policy, consult the UMBC Student Handbook, the Faculty Handbook, or the UMBC Policies section of the UMBC Directory. Resources for students about academic integrity at UMBC are available at <https://academicconduct.umbc.edu/resources-for-students/>.

- All work in homework or an exam must be your own; collaborating on an exam is *not* permitted. Discussions with other students on homework problems are allowed, but you should present your own work in the final turn-in; simply copying other people’s work is violation of academic integrity code.
- If you wish to contest a graded exam, you must make an appeal within *one week* of the return date to the class. End of the semester appeals for earlier exams will be ignored.